

Survival of Hurricane-Generated Coral Fragments and a Disturbance Model of Reef Calcification/Growth Rates

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Summary. Hurricane Gerta, with winds reaching 150 km/h, crossed the Belize barrier reef on September 18, 1978. Breakage and scouring of corals occurred in all zones of the reef to a depth of approximately 25 m. Survivorship of storm-generated coral fragments and detached colonies is strongly size dependent, conforming to the power function $Y = 4.44X^{0.66}$ where Y is the percent of fragments and X is the fragment size. Forty-six percent of detached *Acropora palmata* branches, which are larger ($\bar{X} = 37.6$ cm long) than fragments of other species ($\bar{X} = 16.7$ cm long), survived. Overall, 39% of fragments and detached colonies survived. This high survivorship, which probably increased the total number of colonies present, and redistribution of corals may explain the rapid recovery of reefs from all but the severest hurricanes. Storms appear to prevent coral reefs from reaching a mature state characterized by low calcification and growth rates. Therefore, we suggest that long-term reef calcification and growth rates are highest on reefs periodically disturbed by storms of intermediate intensity.

Adey (1978) proposed a reef morphogenesis model relating reef-building to 1) antecedent topography, 2) growth potential of dominant calcifying organisms relative to wave energy, and 3) to rate of sea level rise. We here propose a model which complements the second component in Adey's model. He proposed that coral growth rates are greatest under normal conditions of moderate wave energy. Combining the two models leads to the prediction that reefs exposed to moderate wave energy and periodic storms of low to moderate intensity should have the highest calcification rates and growth potential.

As a basis for our model, we present data on the size distribution and size-dependent survivorship of hurricane-generated coral fragments. To our knowledge, data of this type have not been included in previous accounts of storm damage to coral reefs (e.g. Stoddart 1963; Glynn et al. 1965; Ball et al. 1967; Perkins and Enos 1968; Maragos et al. 1973).

Introduction

"The coral zoophyte may be levelled by transported masses swept over by the waves; yet like the trodden sod, it sprouts again, and continues to grow and flourish as before."

J.D. Dana, 1853

The luxuriance of life forms on coral reefs is often attributed to relatively benign tropical environments which permit the evolution of numerous species through symbiotic, competitive, or predator-prey type interactions with other reef residents, i.e. coral reef communities are assumed by many to be biologically accommodated (e.g. Sanders 1968; Newell 1971). In fact, the majority of reefs are situated within areas of frequent hurricane activity (Stoddart 1971) and Connell (1978) recently proposed that periodic storms of intermediate intensity are of primary importance in maintaining high diversity on reefs. Our observations on the effects of Hurricane Gerta on Belize Barrier Reef corals suggest that such disturbances may also have a positive effect on long-term reef calcification and growth rates.

Methods

This study was conducted at the Smithsonian Institution's IMSWE Laboratory, Carrie Bow Cay, Belize (~17° N; 88° W), situated on the Belize Barrier Reef. The reef parallels the mainland in a North-South direction and is separated from the coast by a lagoon approximately 25 km across (James et al. 1976). Hurricane Gerta struck the barrier reef with 150 km/h winds on 18 September 1978; the eye crossing the reef 6 km north of Carrie Bow Cay, heading WNW (285°; Rützler 1978). Our observations were made approximately 4 months later, in early January 1979.

In the buttress zone and the upper spur and groove zone (Fig. 1), coral colonies and fragments broken loose from the substratum by the storm were measured along their greatest dimension. Starting at the outer end of each groove, a 50 m transect line was strung shoreward along the center of 5 grooves. All fragments within 1 m of the transect line ≥ 10 cm in greatest dimension were measured, and the presence or absence of live coral tissue recorded. Fragments < 10 cm long were too numerous to measure. Observations were recorded separately for seaward and shoreward halves of the transects. Fragments with worn, eroded calices were considered present prior to Hurricane Gerta and not measured. Because of the abundance of *Acropora palmata*, a large branching coral, in this zone (Fig. 1; James et al. 1976), measurements were listed separately for this species. We did not attempt to identify other coral fragments but they included *Agaricia agaricites*, *Meandrina* sp., *Porites astreoides*, *P. porites*, *Montastrea annularis*, *M. cavernosa*, *Dichocoenia stokesii* and *Acropora cervicornis*.

We also collected, measured, air dried, and weighed all fragments in 10 quadrats 0.25 m² on a rubble bank (Fig. 2.1) in the rubble and pavement zone (Fig. 1).

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TRANSECT ACROSS BARRIER REEF AT CARRIE BOW CAY, BELIZE

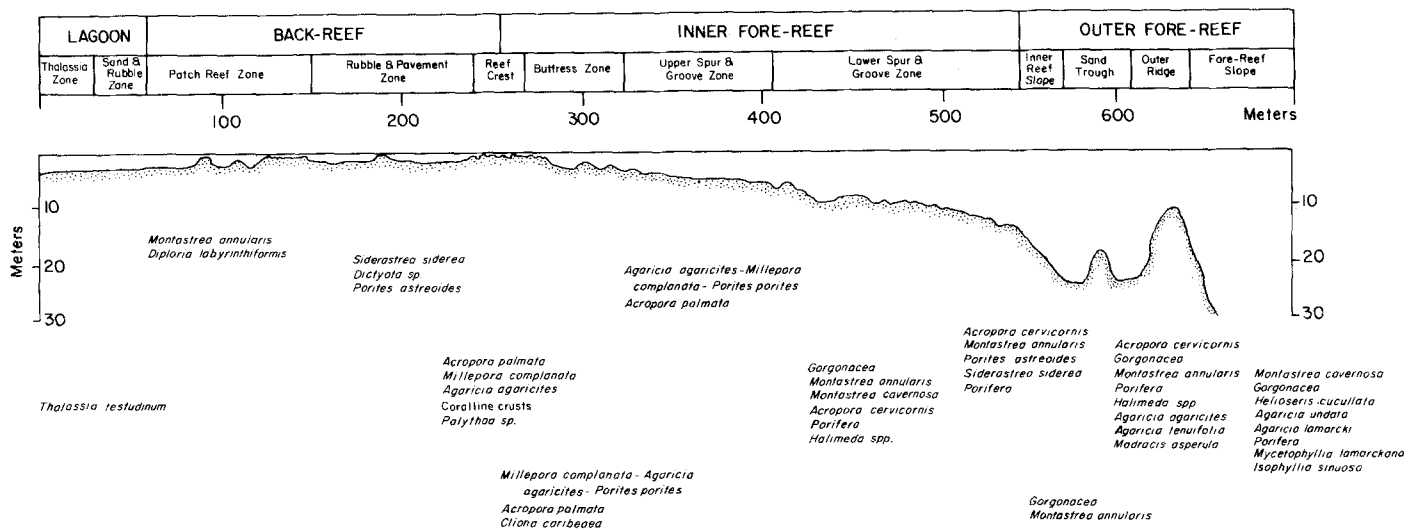


Fig. 1. Cross section of the Belize barrier reef, (after Rützler and Macintyre, in press)

Results

General Observations

Hurricane damage was evident in all zones of the reef except at depths greater than about 25 m on the steep fore-reef slope (Fig. 1). In the back-reef area, many corals were extensively scoured, especially on their upper surfaces (Fig. 2.2). There were also a number of detached, often overturned, massive corals (Fig. 2.3) deposited in this area. In both cases, portions of colonies usually survived. Thus, partial colony mortality due to scour was the major type of damage in this zone.

In the upper spur and groove zone, many corals on the spurs were broken off resulting in accumulation of loose corals, coral branches, and rubble in the grooves (Figs. 2.4, 2.5, 2.6). Data on the size frequency distribution and survivorship of detached corals in the grooves are presented in the next section.

The effects of the storm on the inner reef slope were quite variable. This zone is dominated by *Acropora cervicornis* with massive corals, gorgonians and sponges also abundant on the crest of the slope (Figs. 1 and 2.7). In some locations, nearly total mortality or removal of the biota (Fig. 2.8) was interspersed with areas of minimal disturbance only a few meters away. Typically, many gorgonians and sponges were removed or tattered and corals were scoured (Fig. 3.1). The *A. cervicornis* thicket on the slope was more or less flattened (Fig. 3.2) but live tissue was still present on many broken branches.

On the outer ridge, damage was again quite variable but generally comparable to that on the inner reef slope (Figs. 3.3, 3.4, 3.5, 3.6).

Size and Survival of Coral Fragments

Small fragments were more abundant than large fragments (Fig. 4), possibly reflecting the size distribution of corals present before Gerta and/or indicating that small corals or branches are more susceptible to breakage.

The percent survivorship is highly correlated with fragment size (Fig. 4; $r^2=0.91$, $P<0.001$). Over 50% of fragments larger

than 40 cm survived and 80% or more of fragments ≥ 75 cm survived. Fragments larger than 102 cm were lumped and averaged in the last point because of the low number of fragments in each of these large classes. Survivorship is fit by the power curve $Y = 4.44X^{0.66}$ where Y is the percent of fragments surviving and X is the fragment size in cm. This curve intercepts 100% survival at about 112 cm. Only 2% of the fragments were larger than 112 cm and some were killed indicating the actual curve should asymptote below 100%.

Fragments with surviving coral tissue were significantly larger than those without live tissue (Table 1; $t=8.36$, $P<0.00005$). This relationship holds for *Acropora palmata* branches (Table 1; $t=6.7$, $P<0.00005$), and species in the "Other" category (Table 1; $t=3.4$, $P=0.001$) when considered separately. Thus, the overall result is general rather than due only to the influence of *A. palmata* fragments which are significantly larger than "Other" fragments ($t=16.3$, $P<0.00005$). In addition to the high percentage of *A. palmata* fragments surviving, the bases of most colonies appeared to survive and were in the process of healing branch stumps (Fig. 3.7).

The abundance of *Acropora palmata* in the upper spur and groove zone increases toward the reef crest (Fig. 1). More *A. palmata* fragments were found in the shoreward transects than in the seaward transects (Table 2; $P<0.06$; Mann-Whitney U-test), suggesting little net seaward transport. Thus, *A. palmata* fragments tend to remain in the reef zone inhabited by the parent colonies.

Rubble Bank Fragments

The fragments ($N=1507$) in the rubble bank had a mean length of 5.5 cm (s.d.=2.4) and a mean weight of 33.7 g (s.d.=65.9). Fragment weight (Y) is correlated with fragment length (X; $r=0.69$, $P<0.001$) and gives the regression $Y = -69.3 + 18.78 X$. This regression may underestimate the weight of large fragments such as *Acropora palmata*. A 25 cm branch of *A. palmata* weighed 3 times the predicted weight, a discrepancy probably due to the high bulk density ($\sim 2.0 \text{ g} \cdot \text{cm}^{-3}$; Chamberlain 1978; Highsmith

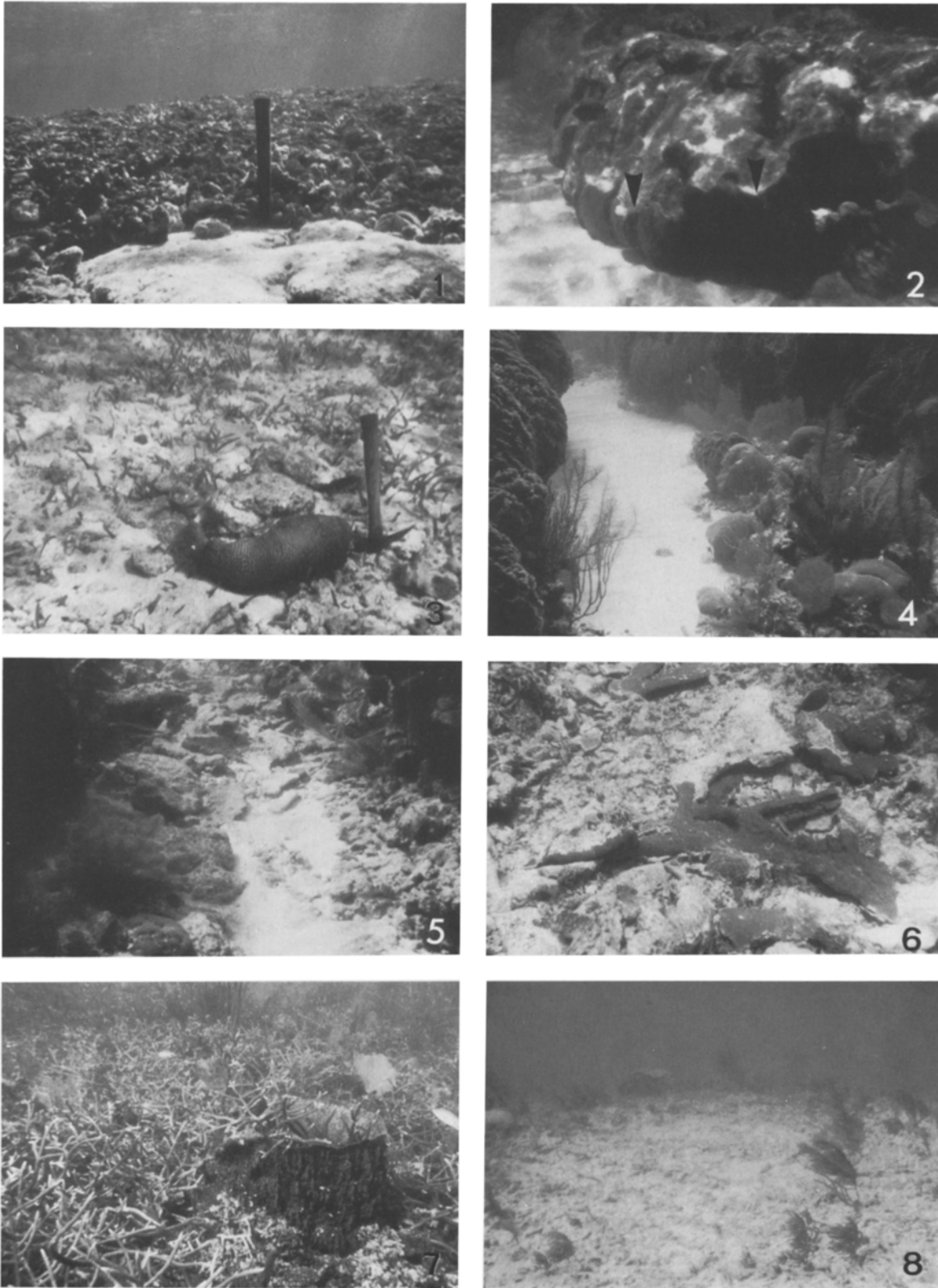


Fig. 2.1. Rubble bank in the rubble and pavement zone. Hammer is 30 cm long. **2.** Severely scoured *Montastrea annularis* colony in the patch reef zone of the back-reef. Only tissue on the lower part of the head survived (arrows). **3.** Overturned but living head of *Diploria* sp. transported to the rubble and pavement zone, probably from the reef crest. **4.** Groove between two spurs of *Agaricia agaricites* before (April, 1978) Hurricane Gerta. **5.** Rubble-strewn groove following Hurricane Gerta. **6.** Several live *Acropora palmata* fragments and 1 live *Porites astreoides* in a groove after Hurricane Gerta. **7.** Crest of inner reef slope before (April, 1978) Hurricane Gerta. Dominant coral is *Acropora cervicornis*. **8.** Area on crest of inner reef slope devastated, possibly by a breaking wave, during Hurricane Gerta



Fig. 3.1. Typical scene on upper margin of inner reef slope after Hurricane Gerta. **2.** Thicket of *Acropora cervicornis* flattened by hurricane. Compare with Fig. 2.7. **3, 4.** Crest of outer ridge before (April, 1978) Hurricane Gerta. **5, 6.** Hurricane damage to crest of outer ridge. Compare with Fig. 3.3, 3.4. **7.** Live base of *Acropora palmata* colony from which all branches were broken. **8.** Enewetak table acroporan killed during a storm (Sept. 1972) provides substrate for coral recruitment (photographed Sept. 1977)

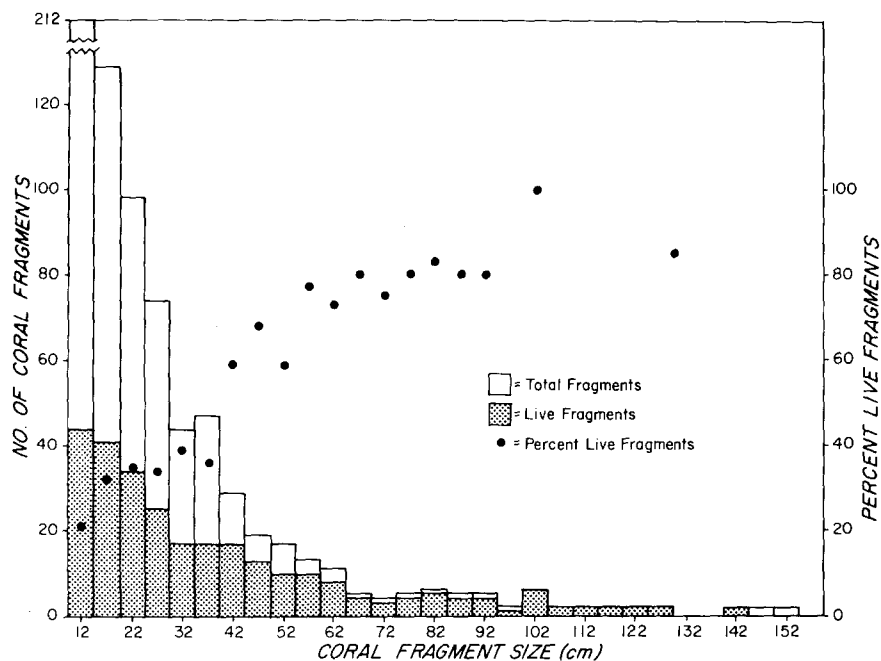


Fig. 4. Size distribution of coral fragments and detached colonies in grooves following Hurricane Gerta

Table 1. Mean size of coral fragments and detached coral colonies in the upper spur and groove zone (Fig. 1) following Hurricane Gerta

	$\bar{X}$ (cm)	S.D.	N	% live
All fragments:				
Live	37.0	26.6	289	
Dead	22.6	15.5	466	
Combined	28.3	20.0	750	39
<i>Acropora palmata</i> :				
Live	46.3	27.7	189	
Dead	30.3	18.7	223	
Combined	37.6	24.6	412	46
Other corals:				
Live	19.5	10.9	100	
Dead	15.6	5.9	243	
Combined	16.7	7.9	343	29

Table 2. Number of *Acropora palmata* fragments in 25 m seaward and shoreward transects

Transect No.	No. seaward	No. shoreward
1	20	167
2	4	22
3	14	29
4	13	22
5	102	20

1979a; cf. 2.94 for solid aragonite) and robust branches of this species (cf. Figs. 2.1 and 2.6).

Much of the material in the rubble bank was probably rubble before Hurricane Gerta judging by algal encrustment and calice erosion. New rubble included some live *Porites porites* branches.

Discussion

The small fragments in the rubble bank give an indication of the upper size of particles suspended in great quantity during moderate hurricanes. Larger fragments may be transported in quantity during more intense storms (Maragos et al. 1973; Hernandez-Avila et al. 1977). Movement of these particles, combined with sand, must play an important role in scouring and breaking reef corals. On the other hand, the large, heavy *A. palmata* branches were most abundant in the shoreward 25 m of the transects (Table 2) where *A. palmata* colonies were most common, suggesting that, although *A. palmata* is an important reef-forming coral (e.g. Goreau 1959; Shinn et al. 1977; Lighty 1977; Adey 1978), broken branches are not generally transported to deeper reef zones and thus contribute mostly to formation of the upper few meters of the barrier reef, as found for Holocene reefs at Curacao and Bonaire (Focke 1978) and Pleistocene reefs at Barbados (Mesolella 1967). During periods of sea level rise, *A. palmata* could of course contribute to a much larger portion of the reef frame (Macintyre and Glynn 1976).

A surprisingly large proportion of fragments generated by Hurricane Gerta survived (Fig. 4; Table 1). Most small fragments including those less than 10 cm long were blades broken from large *Agaricia agaricites* colonies, probably the dominant spur-forming coral in the upper spur and groove zone (Fig. 1; James et al. 1976). The highly significant correlation between fragment size and survival (Fig. 4) suggests that rapid coral growth rates may, among other things, be an adaptation to periodic disturbances.

Acropora palmata fragments, with nearly 50% survivorship, were larger than fragments of other species (Table 1). We attribute this to the high growth rate, skeletal strength, and growth form of *A. palmata*. The skeleton is strong, probably a necessity to support long branches (Chamberlain, 1978), because of its relatively high bulk density (see results) and because the CaCO_3 crystals are oriented approximately perpendicular to the direction of stress, i.e. parallel with the long axis of the branch (Chamberlain 1978). Consequently, intact branches tend to break off near the base

where mechanical stresses due to storm currents and suspended rubble are greatest.

A. palmata branch growth rates are typically 7 to 8 cm·yr⁻¹ (Bak 1976; Gladfelter et al. 1978). Thus, most *A. palmata* branches can reach the 80% survival size (Fig. 4) in a period of approximately 10 years – the average hurricane frequency at Belize (Stoddart 1963). The periodic detachment and regeneration of branches, with a high percentage of the detached branches surviving, may be the predominant mode of reproduction and distribution for *A. palmata* in Belize. Based on low numbers of juveniles and the frequent development of colonies from fragments, Bak and Engel (1979) reached a similar conclusion for *A. palmata* in Curaçao and Bonaire. To what extent, if any, *A. palmata* is dependent on storms for its abundance and local distribution is an open question. Hurricanes of low to moderate intensity appear to be disasters (*sensu* Harper, 1977) in this context: “A disaster recurs frequently enough for there to be reasonable expectation of occurrence within the life cycles of successive generations – e.g. hurricanes at ca. 70 year intervals are ‘disasters’ in the life of a forest tree and the selective consequence may be expected to leave relevant genetic and evolutionary memories in succeeding generations. A ‘catastrophe’ occurs sufficiently rarely that few of its selective consequences are relevant to the fitness of succeeding generations. The selective consequence of disasters is therefore likely to be to increase short-term fitness and the consequence of catastrophes is to decrease it.” Very intense hurricanes, e.g. Hattie in 1961 (Stoddart 1963, 1974), probably constitute catastrophes for corals in the center of the storm track but disasters for corals located along the margin. In contrast to disasters, recovery from catastrophes, e.g. the central path of major hurricanes, severe *Acanthaster* plagues, or episodic pollution, will depend on recruitment by larvae from a distant source and thus be much slower (e.g. Stoddart 1974).

For the years 1900 to 1962, 286 hurricanes and 212 tropical storms of less severity have been recorded in the Caribbean (Glynn et al. 1965). To the extent that these data are representative of storm frequency over geologic time, storms must surely be an important selective factor in the evolution of hermatypic corals (Highsmith 1980). *Acropora palmata*, by virtue of its high growth rate, colony shape, and skeletal features is extremely well adapted to the periodic disturbance regime at Belize, probably accounting for the great success of this branching coral in a reef zone often considered the domain of calcareous algae and massive corals (e.g. Darwin 1874; Wells 1954; Stoddart 1969; Braithwaite 1971; Barnes 1973; Vosburgh 1977). For hermatypic corals in general, the fragmentation of colonies appears to be a widespread and extremely important method of reproduction and distribution (Highsmith 1980).

Similar comments may apply to the staghorn coral, *Acropora cervicornis*, from which *A. palmata* apparently evolved (Frost 1977). Time did not permit transecting on the inner reef slope and inner slope of the outer ridge (Fig. 1), areas dominated by *A. cervicornis*. Although the staghorn thickets were flattened, it was obvious that many fragments survived. *A. cervicornis* also has high growth rates (Goreau and Goreau 1959; Shinn 1966; Glynn 1973; Gladfelter et al. 1978) and a very dense skeleton (bulk density ~2.4; Highsmith 1979a). For both species, dense skeletons may inhibit breakage into small pieces and also reduce transport to less favorable reef zones.

Among other things, coral skeletons are often thought to provide protection from predators because each polyp is situated within a skeletal calice into which it can withdraw (e.g. Glynn 1973; Vermeij 1978). An extremely important aspect of storm

damage to reefs, however, is the great amount of coral tissue destroyed by scouring. The role of calice size and architecture in supporting and protecting tissue during storms should be investigated.

If killed, coral skeletons will contribute to reef growth in that large portions of reefs are composed of lithified sediments (Dana 1853; Newell 1971; Weber and Woodhead 1979), and they provide settlement sites for coral larvae (Fig. 3.8); Stoddart 1974). Gorgonians and sponges also suffer considerable detachment and mortality in storms, freeing space for recruitment and/or growth of corals and calcareous algae.

Most reef corals have indeterminate growth (Buddemeier and Kinzie, 1976; Highsmith 1979b) but may still have low growth rates due to limited substrate, crowding, or proximity to the water surface (see Fig. 8 in Goreau 1959; plate 6 in Storr, 1964; plates IC and VA in Milliman 1973; Figs. 4 and 5 in Scatterday 1974; Macintyre Burke and Stuckenrath 1977; ref. 34 in Adey 1978). Among corals that do have determinate growth are two species, *Pocillopora damicornis* and *Stylophora pistillata*, that are abundant on reefs and have high growth rates that decline as the colonies become relatively large (Maragos 1972; Loya 1976). Thus, moderate storms may have the net effect of thinning crowded or slow-growing colonies, distributing corals to somewhat deeper of soft-bottom locations where larvae are unable to settle (Highsmith 1980), providing settlement sites for larvae, and initiating regenerative coral growth which, in at least some species, is more rapid than normal growth (Wood-Jones 1907; Stephenson and Stephenson 1933; Maragos 1972; Loya 1976). Therefore, long-term reef calcification rates may be higher in locations with periodic disturbances of low to moderate intensity.

A Disturbance Model of Reef Calcification/Growth Rates

If long-term calcification rates are increased by certain types of disturbances as suggested above, rates on undisturbed reefs must decline as they “mature” (Fig. 5). We define “mature reefs” as those in which most hard substrate is occupied, resulting in crowding, space competition, and low recruitment success. Also, large areas of such reefs have approached the water surface or are even emergent, so that further upward growth by calcifying organisms is inhibited (Goreau 1959; Pichon 1974; Scatterday 1974; Macintyre and Glynn 1976; Shinn 1963; Frost 1977; Macintyre, Burke and Stuckenrath 1977; ref. 34 in Adey 1978). According to the model (Fig. 5), calcification rates on “mature reefs” will shift from zone B₂ to zone C following storms of low to moderate intensity or to zone A following a major hurricane. Reefs with calcification rates in zone C will experience a decline in rates to zone A if struck by a major hurricane. For reefs with pre-disturbance calcification rates in zone A, any disturbance resulting in mortality of important calcifying organisms will have a negative effect on calcification rates. We have shown broad overlapping zones rather than specific rates or zones with sharp edges because not enough data are available to pinpoint these rates and because the rates are relative to the maximum potential calcification rate in the region being considered (Adey 1978).

Since reef calcification rate is an important factor in reef growth rates, this model, after adjustment for CaCO₃ export, can also be used to characterize reef development rates. Again, the data available do not permit inclusion of precise rates in the model. However, maximum long-term, i.e. >1,000 years, vertical growth rates of coral reefs seldom exceed 10 m/1,000 yrs (Table 3). Occasionally, reefs have very high growth rates for short

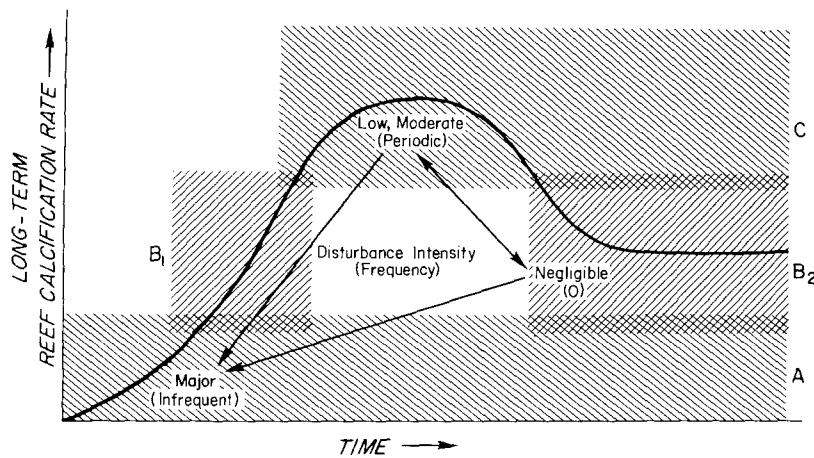


Fig. 5. Disturbance model of coral reef calcification and growth rates. See text. Rate zone *A*: coral communities, reefs affected by catastrophes. Rate zone *B*₁: young reefs, reefs recovering from catastrophes (several years after the catastrophe). Rate zone *B*₂: mature, undisturbed reefs. Rate zone *C*: rapidly developing reefs prevented from maturing (zone *B*₂) by periodic disturbances of intermediate intensity

Table 3. Vertical growth rates of coral reefs

Location	Growth rate (m/1,000 yrs)		Source
	$\bar{X}$	Range	
Caribbean:			
Belize	1.1–5.0	0.7– 7.8	Purdy 1974
Alacran Reef, Mexico		2.0–12.0	Macintyre et al. 1977
Florida		0.4– 4.9	Shinn et al. 1977
Florida		9.0	Lighty 1977
St. Croix		0.2– 8.0	Adey et al. 1977
St. Croix		1.0–15.0	Adey 1975
Jamaica		1.2	Goreau and Land 1974
Jamaica		0.6– 1.0	Land 1974
Panama	0.6–3.9	0.4–10.8	Macintyre and Glynn 1976
Indo-Pacific:			
Bikini, Enewetak		1.0– 2.0	Tracey and Ladd 1974
Great Barrier Reef		3.0	Davies et al. 1977
New Guinea	< 4.7	~8.0max.	Chappell and Polach 1976
Reunion Island	4.0	1.0–10.0	Montaggioni 1976
Hawaii		1.0– 2.0	Easton and Olson 1976
Mururoa, Tuamotu Islands		0.2– 3.4	Lalou et al 1966
Panama	1.3–7.5	0.5–20.8	Glynn and Macintyre 1977

periods of time (~200–800 years) due to the predominance of branching corals with high calcification rates. Following the period of very rapid accretion, very little further growth occurs as the reef approaches sea level, i.e. matures (Fig. 5; e.g. Macintyre and Glynn 1976; Glynn and Macintyre 1977; Macintyre et al. 1977). Such reefs may not be terribly stable over geologic time because they consist of a meshwork of branches rather than solid frame (see Fig. 6 in Glynn et al. 1972; Fig. 2; in Glynn and Macintyre 1977).

Holocene reef development reaches its apex in the Caribbean (Adey 1978) where an average of 4.5 hurricanes and 3.4 tropical storms occur each year (Glynn et al. 1965). The net effect of frequent storms may be to maintain many Caribbean reefs in zone *C* of the model.

Conclusions

The relatively high survivorship of storm-generated coral fragments, especially *Acropora palmata*, may explain the surprisingly rapid recovery of coral reefs from all but the severest hurricanes (Stoddart 1974; Shinn 1976). The appearance of great damage is due to the large number of breakage scars, disarray of fragments, and reduced live coral tissue cover due to scour. Although percent live cover and mean colony size are reduced, the total number of coral colonies on reefs may actually be increased by hurricanes of moderate intensity, such as Gerta, via fragmentation of colonies. Most corals were scoured but seldom completely killed. Thus, much of the recovery process simply involves growth of surviving coral tissues over skeletal areas killed during the storm.

The major effect of periodic storms of intermediate intensity may be to condense or export CaCO_3 and to redistribute corals so that continued growth is not limited by crowding or proximity to sea surface. Consequently, as a complement to the reef morphogenesis model of Adey (1978), we have proposed a model which predicts that coral reefs so disturbed will have the highest long-term calcification and growth rates.

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